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B.TECH
(SEM V) THEORY EXAMINATION 2022-23
MASS TRANSFER-I

Time: 3 Hours

Total Marks: 100

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.
2. Any special paper specific instruction.

SECTION A

1. Attempt all questions in brief.

2 x 10 = 20

- (a) What are different methods of conducting mass transfer operations at commercial scale?
- (b) How the eddy diffusion is differentiated from molecular diffusion?
- (c) What is the effect of pressure and temperature on absorption?
- (d) State different types of packing materials and explain their selection criterion.
- (e) Compare N type flux and J type flux.
- (f) Write applications of rotary dryer.
- (g) Explain when Knudsen diffusion occurs in solids.
- (h) Define HTU and NTU for packed tower.
- (i) What is the significance of an operating line?
- (j) What is crystallization?

SECTION B

2. Attempt any three of the following:

10 x 3 = 30

- (a) State and explain Fick's first law of diffusion. Also, differentiate between flux N_A and J_A . Additionally, Prove that for unidirectional binary diffusion $J_A = -J_B$.
- (b) Explain in detail about choice of solvent used for absorption and absorption with chemical reaction.
- (c) Discuss classification of mass transfer operation in detail.
- (d) Atmospheric air at 1.0132 bar has a DBT of 30°C and a WBT of 25°C. Compute i) The partial pressure of water vapour ii) The specific humidity iii) The dew point temperature iv) The relative humidity v) The degree of saturation.
- (e) Write the different types of crystallizer used in industry. With the neat sketch explain working of Forced-circulation crystallizer.

SECTION C

3. Attempt any one part of the following:

10 x 1 = 10

- (a) Discuss the penetration theory of diffusion.
- (b) Consider the following two cases for a binary mixture of ideal gases A and B under steady state conditions. In Case 1, the diffusion of A occurs through non-diffusing B. In Case 2, equimolar counter diffusion of A and B occurs. In both the cases, the total pressure is 100 kPa and the partial pressures of A at two points separated by a distance of 10 mm are 10 kPa and 5 kPa. Assume that the Fick's first law of diffusion is applicable. What is the ratio of molar flux of A in Case 1 to that in Case 2?

4. Attempt any *one* part of the following: 10 x 1 = 10
- (a) Derive Kremser equation for the continuous counter-current extraction operation.
 - (b) A countercurrent absorption tower is designed to remove 95% of component A from an incoming binary gas mixture using pure solvent B. The mole ratio of A in the inlet gas is 0.02. The carrier gas flow rate is 50 kmol/h. The equilibrium relations is given by $Y=2X$, where Y and X are the mole ratios of A in the gas and liquid phases, respectively. If the tower is operated at twice the minimum solvent flow rate, calculate the mole ratio of A in the exit liquid stream ?
5. Attempt any *one* part of the following: 10 x 1 = 10
- (a) Give the classification of cooling tower based on air drift. Explain the construction and operation of a forced draft cooling tower with neat sketch.
 - (b) An air-water vapor mixture has a dry bulb temperature of 60°C and a dew point temperature of 40°C. The total pressure is 101.3 kPa and the vapor pressures of water at 40°C and 60°C are 7.30 kPa and 19.91 kPa, respectively. Calculate (i) The humidity of air sample expressed as kg of water vapor / kg of dry air (ii) The wet bulb temperature T_w for the above mixture ?
6. Attempt any *one* part of the following: 10 x 1 = 10
- (a) With a neat sketch, explain drying curve.
 - (b) It takes 6 hours to dry a wet solid from 50% moisture content to the critical moisture content of 15%. How much longer will it take to dry the solid to 10 % moisture content, under the same drying conditions (the equilibrium moisture content of the solid is 5%) ?
7. Attempt any *one* part of the following: 10 x 1 = 10
- (a) A batch of 1000 kg of KCl is dissolved in sufficient water to make a saturation solution at 363 K where the solubility is 35 wt % KCl in water. The solution is cooled to 293 K, at which the temperature its solubility is 25.4 wt %.
 - (i) What is the weight of water required for solution and the weight of crystals of KCl obtained?
 - (ii) What is the weight of crystals obtained if 5 % of the original water evaporates on cooling?
 - (b) Write the short note of the following-
 - i. Super saturation
 - ii. McCabe 'Δ Law'
 - iii. Factors governing nucleation
 - iv. Diffusion reaction theory of crystal growth.